

PREFACE FOR FACULTY

The Window Method is the classical analytical approach to FIR filter design. Starting from an ideal brick-wall frequency response $H_d(e^{j\omega})$, its infinite non-causal impulse response $h_d[n]$ is truncated and smoothed using a finite window function $w[n]$.

Pedagogical Strategy:

1. Derive the ideal lowpass impulse response: $h_d[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$ where $\tau = \frac{N-1}{2}$.
 2. Explain the **Gibbs Phenomenon**: Truncating with a rectangular window causes 8.95% peak overshoot at band edges, which does not vanish even as $N \rightarrow \infty$.
 3. Analyze the fundamental trade-off: **Mainlobe width (transition bandwidth) vs. Sidelobe level (stopband attenuation)**.
 4. Define and analyze Rectangular, Bartlett (triangular), and Hann (Hanning) windows.
 5. Complete step-by-step filter design numericals.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** ideal impulse responses $h_d[n]$ for Lowpass, Highpass, Bandpass, and Bandstop filters.
 2. **Explain** the origins and mathematical properties of the Gibbs phenomenon.
 3. **Design** linear-phase FIR filters using Rectangular, Bartlett, and Hann windows to meet transition bandwidth and attenuation specs.
 4. **Evaluate** the frequency spectrum of window functions.
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2. MATHEMATICAL FOUNDATIONS

2.1 Ideal Filter Impulse Responses ($\tau = \frac{N-1}{2}$)

- **Lowpass (ω_c):**

$$h_{d,LP}[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$$

- **Highpass (ω_c):**

$$h_{d,HP}[n] = \delta[n-\tau] - \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$$

- **Bandpass (ω_{c1}, ω_{c2}):**

$$h_{d, \text{BP}}[n] = \frac{\sin(\omega_{c2}(n - \tau))}{\pi(n - \tau)} - \frac{\sin(\omega_{c1}(n - \tau))}{\pi(n - \tau)}$$

2.2 Window Functions Formulation ($0 \leq n \leq N - 1$)

1. Rectangular Window:

$$w[n] = 1, \quad \Delta\omega = \frac{4\pi}{N}, \quad \text{Peak Sidelobe: } -13 \text{ dB}, \quad A_s = 21 \text{ dB}$$

2. Bartlett (Triangular) Window:

$$w[n] = 1 - \frac{2|n - \tau|}{N - 1}, \quad \Delta\omega = \frac{8\pi}{N}, \quad \text{Peak Sidelobe: } -25 \text{ dB}, \quad A_s = 25 \text{ dB}$$

3. Hann (Hanning) Window:

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N - 1}\right), \quad \Delta\omega = \frac{8\pi}{N}, \quad \text{Peak Sidelobe: } -31 \text{ dB}, \quad A_s = 44 \text{ dB}$$

3. WORKED NUMERICAL EXAMPLES

Example 22.1: FIR Lowpass Filter Design using Hann Window

Problem: Design a causal FIR lowpass filter with cutoff $\omega_c = \frac{\pi}{4}$ rad/sample and length $N = 7$ using a Hann window.

Solution:

- Length $N = 7 \implies \tau = \frac{7-1}{2} = 3$.
- Ideal impulse response:

$$h_d[n] = \frac{\sin(\frac{\pi}{4}(n - 3))}{\pi(n - 3)}$$

- $h_d[3] = \frac{\omega_c}{\pi} = \frac{\pi/4}{\pi} = 0.2500$
 - $h_d[2] = h_d[4] = \frac{\sin(-\pi/4)}{-\pi} = \frac{\sqrt{2}/2}{\pi} = \frac{0.7071}{\pi} = 0.2251$
 - $h_d[1] = h_d[5] = \frac{\sin(-2\pi/4)}{-2\pi} = \frac{1}{2\pi} = 0.1592$
 - $h_d[0] = h_d[6] = \frac{\sin(-3\pi/4)}{-3\pi} = \frac{0.7071}{3\pi} = 0.0750$
- Hann Window Values ($w[n] = 0.5 - 0.5 \cos(\frac{2\pi n}{6})$):**
 - $w[3] = 0.5 - 0.5 \cos(\pi) = 1.0000$
 - $w[2] = w[4] = 0.5 - 0.5 \cos(2\pi/3) = 0.5 - 0.5(-0.5) = 0.7500$
 - $w[1] = w[5] = 0.5 - 0.5 \cos(\pi/3) = 0.5 - 0.5(0.5) = 0.2500$
 - $w[0] = w[6] = 0.5 - 0.5 \cos(0) = 0.0000$
- Final FIR Filter Coefficients ($h[n] = h_d[n] \cdot w[n]$):**
 - $h[0] = h[6] = 0.0750 \times 0.0000 = 0.0000$
 - $h[1] = h[5] = 0.1592 \times 0.2500 = 0.0398$
 - $h[2] = h[4] = 0.2251 \times 0.7500 = 0.1688$
 - $h[3] = 0.2500 \times 1.0000 = 0.2500$

$$h[n] = 0.0000, 0.0398, 0.1688, 0.2500, 0.1688, 0.0398, 0.0000$$

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) What is the Gibbs phenomenon? Explain how window functions reduce Gibbs ringing at the expense of transition bandwidth. (6 Marks) **(b)** Design a 9-tap FIR Highpass filter with cutoff $\omega_c = \frac{\pi}{3}$ using a Bartlett window. (9 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Explanation of 8.95% overshoot at step discontinuities and Fourier truncation (3 Marks)
 - Tapering effect of windows: Smooth boundary transitions suppress high-frequency sidelobes but widen mainlobe transition band (3 Marks)
- **Part (b):**
 - $N = 9, \tau = 4$.
 - $h_d[n] = \delta[n - 4] - \frac{\sin(\pi(n-4)/3)}{\pi(n-4)}$ (3 Marks)
 - Bartlett window: $w[n] = 1 - \frac{|n-4|}{4}$ for $n = 0, \dots, 8$ (2 Marks)
 - Evaluation of all 9 coefficients $h[n] = h_d[n]w[n]$ with symmetry (4 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

N = 7
tau = 3
wc = np.pi / 4
n = np.arange(N)

# Ideal LPF
hd = np.zeros(N)
hd[n != tau] = np.sin(wc * (n[n != tau] - tau)) / (np.pi * (n[n != tau] - tau))
hd[tau] = wc / np.pi

# Hann window
w = 0.5 - 0.5 * np.cos(2 * np.pi * n / (N - 1))
h = hd * w

print("Designed FIR Filter Coefficients h[n]:")
print(np.round(h, 4))
```